CHAPTER 9 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

UAS Weaponization Risks and Counter-UAS Security Implications

UAS Weaponization: Definitions and Security Context

What counts as weaponization

A UAS is weaponized when someone adapts or employs it to cause harm. The same aircraft class may also support legitimate work.

What operators can observe

Location, flight behavior, and available identity data help characterize an event. Record uncertainty about intent and authorization.

What the assessment supports

Assess potential consequences for people, airspace, and essential services before escalating.



Internet of Battlefield Things Drone Attacks and Defensive Packet Sniffing

Networked mission equipment

A battlefield Internet of Things may connect aircraft, sensors, field gateways, and operator terminals.

Defensive packet observation

Use authorized captures to compare timing, endpoints, and traffic volume with a known-good baseline.

Limits of the evidence

Encryption can conceal payload content. Correlate unusual traffic with endpoint and gateway logs before drawing conclusions.



Security Aspects of the Internet of Drones Against Drone Attacks

Trust boundaries

Separate aircraft command and control, payload services, and management access.

Identity and message integrity

Authenticate devices and protect command messages against modification and replay.

Degraded operation

Monitor unexpected peers and repeated failures. Test how the fleet handles a lost link in simulation.



Dual-Use UAV Technologies and Misuse Risks

Shared capabilities

Cameras, autonomous navigation, and modular payloads support legitimate inspection, mapping, and emergency response.

Why context matters

Evaluate authorization, location, behavior, and potential consequences when assessing possible misuse.

Controls for authorized use

Limit access to mission software and payloads. Record who approves each mission configuration.



Payload-Based Threats and Mission-Level Risks

Payload assurance

Verify payload identity, configuration, and approval before flight. Document discrepancies.

Mission consequences

Consider how a payload anomaly could affect flight control, collected data, people, or nearby infrastructure.

Escalation criteria

Define when to pause the mission, retain evidence, and refer the event for review.



Critical Infrastructure Protection Against Malicious UAVs

Site assessment

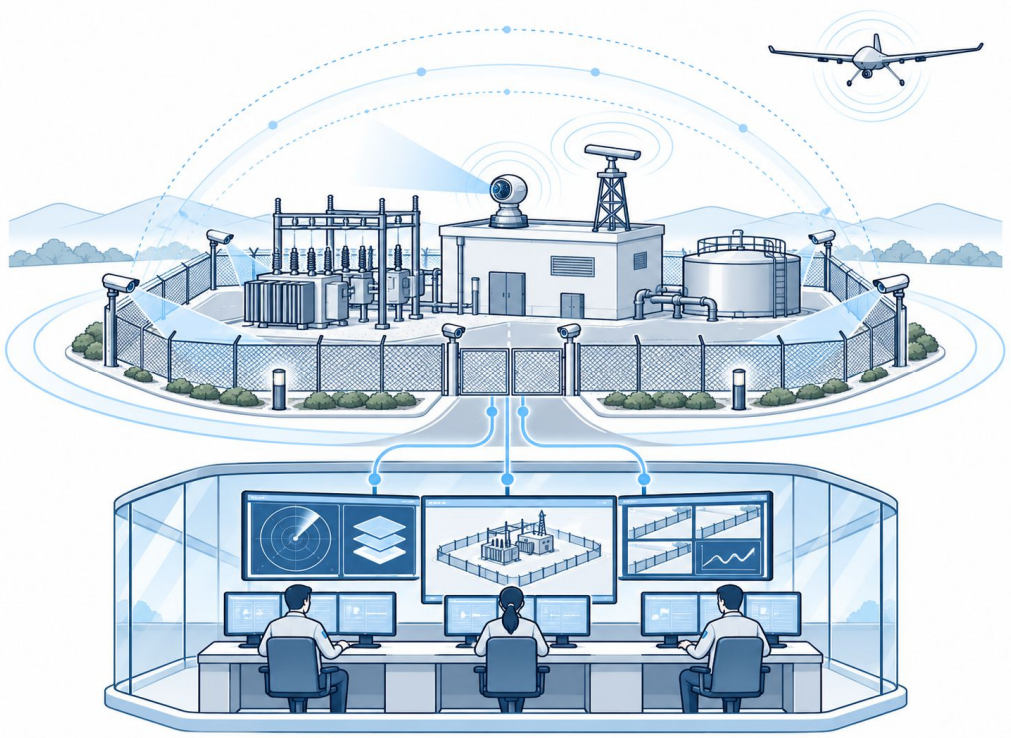
Identify essential assets, normal air activity, and the consequences of a service interruption.

Layered observation

Combine site observations with authorized sensor data to corroborate a track and assess uncertainty.

Response coordination

Define notification routes and responsibilities. Confirm the authority and safety requirements for any proposed response.



Risk Mitigation and Policy Recommendations

Before the operation

Assign decision authority, set escalation criteria, and agree on how to document observations.

During an incident

Record timestamps, sources, confidence, and decisions. Protect evidence and restrict access to it.

After the incident

Review false alarms and response delays. Update procedures and test them in an exercise.



Counter-UAS System Architectures and Operational Accountability

From sensors to tracks

Correlate observations and preserve uncertainty, disagreements, and track history.

From tracks to decisions

Keep detection, identification, authorization checks, and response approval as distinct steps.

Audit trail

Record the evidence, decision-maker, time, and approved action. Apply the site’s retention and access rules.

